

Optical Parameter Estimation for Hair and Fur using Differentiable Rendering

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Figure 1: Rendering results of hair (top) and fur (bottom) using our parameter estimation method. The leftmost images (initial state) are the rendering results with the initial parameter values. Our method iteratively updates the optical parameters using stochastic gradient method with our path tracer. The converged results match the target image taking into account indirect illumination. We develop a differentiable path-tracing renderer tailored to estimate optical parameters of BCSDf. To avoid getting stuck in a local minimum, we propose a two-stage estimation method where the first stage estimates two optical parameters affecting hair and fur’s entire appearance. Then, all the optical parameters are jointly estimated in the second stage. Experimental results substantiate that our method can automatically estimate optical parameters of hair and fur.

ABSTRACT

In this paper, we propose a parameter estimation method for Bidirectional Curve Scattering Distribution Function (BCSDF) for hair and fur using differentiable rendering. Given fiber geometries, illumination, and a target rendered image, our method jointly estimates optical parameters of BCSDF so as to match the target image taking into account indirect illumination. We develop a differentiable path-tracing renderer tailored to estimate optical parameters of BCSDF. To avoid getting stuck in a local minimum, we propose a two-stage estimation method where the first stage estimates two optical parameters affecting hair and fur’s entire appearance. Then, all the optical parameters are jointly estimated in the second stage. Experimental results substantiate that our method can automatically estimate optical parameters of hair and fur.

CCS CONCEPTS

• Computing methodologies → Reflectance modeling.

KEYWORDS

Hair, Fur, BCSDF, Differentiable Rendering

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1 INTRODUCTION

Humans and animals are central characters in movies, commercial films, and games. Rendering such characters often involves a physically-based rendering of hair and fur. Physically-based hair and fur reflectance models [Marschner et al. 2003; Yan et al. 2015] have many optical parameters such as longitudinal and azimuthal roughness of cuticle, absorption coefficients in hair fibers, etc. Obtaining a plausible rendering appearance of hair and fur imposes labor-intensive parameter tweakings on users. We propose an estimation method of optical parameters for physically-based hair and fur reflectance models from a single synthesized image using differentiable rendering to alleviate this problem.

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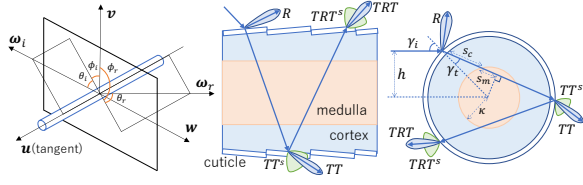


Figure 2: Longitudinal angles θ_i, θ_r and azimuthal angles ϕ_i, ϕ_r for incident and reflected directions ω_i and ω_r .

1.1 Related Work

Zinke et al. proposed a parameter fitting method for hair colors by taking a photograph of a wisp of hair coiled around a cylinder [Zinke et al. 2009]. Bonneel et al. proposed a parameter estimation method of hair appearance by searching features of an input photograph that matches the precomputed dataset [Bonneel et al. 2009]. Yan et al. proposed a reflectance model for hair and fur fibers, referred to as *Double Cylinder Model*, and fitted parameters to the measured fur reflectance profiles [Yan et al. 2017, 2015]. Sun et al. reconstruct geometries of hair fibers using multi-view photometric stereo [Sun et al. 2021]. This method employs the BSDF model proposed by Chiang et al. [2016] and estimates two optical parameters (longitudinal roughness and absorption coefficient in fiber) of the BSDF model. Although this method can reconstruct realistic hair geometries, this method estimates only two optical parameters and approximates the multiple scattering of lights in hair volumes with empirically hair albedo, while our method estimates more parameters and considers the multiple scattering of light within a hair volume using path tracing. Khungurn et al. [2016] proposed a parameter estimation method to match the appearance of fabrics using differentiable rendering, but this method is limited to cloth fibers, while our method handles hair and fur fibers, which have more complex scattering and attenuation behavior of light.

1.2 Background

Our method estimates the optical parameters of BCSDf S that describes the scattering behavior of light on the fiber. As the BCSDf, we employ the double cylinder model proposed by Yan et al. [2017] since this model unifies the reflectance models of hair and fur fibers.

In the double cylinder model, a fiber consists of the cortex, medulla, and cuticle, as shown in Fig. 2(center). The parameter set Θ for the double cylinder model is shown in Table 1. The light scattered by a fiber is decomposed into five paths: R, TT, TRT, TT^s and TRT^s , where R and T denote the reflection and the transmission of light, the last two paths TT^s and TRT^s take into account the scattering in the medulla. The BCSDf S in the double cylinder model is further decomposed into a longitudinal function M_p and an azimuthal function N_p as:

$$S(\theta_i, \theta_r, \phi_i, \phi_r, h) = \frac{\sum_{p \in \mathcal{P}^u} M_p(\theta_i, \theta_r) N_p(\phi, h)}{\cos^2 \theta_i} + M^s(\theta_i, \theta_r, \phi) \frac{\sum_{p \in \mathcal{P}^s} N_p^s(\phi, h)}{\cos^2 \theta_i},$$

where $\mathcal{P}^u = \{R, TT, TRT\}$ is the path set of the unscattered components, $\mathcal{P}^s = \{TT^s, TRT^s\}$ is the path set of the scattered components, and M^s, N_p^s are the longitudinal and azimuthal functions

Table 1: Optical parameters Θ for double cylinder model.

Parameter	Meaning
α	longitudinal shift due to the scale tilt for cuticle
β_m	longitudinal roughness of cuticle
β_n	azimuthal roughness of cuticle
η	refractive index of cortex and medulla
l	layers of cuticle
$\sigma_{c,a}$	absorption coefficient of light in cortex
$\sigma_{m,s}$	scattering coefficient of light in medulla
$\sigma_{m,a}$	absorption coefficient of light in medulla
g	anisotropy factor of scattering in medulla

of the scattered components, ϕ is the relative angle of azimuthal $\phi = \phi_r - \phi_i$. $h \in [-1, 1]$ represents the incident position of the light in the azimuthal cross section as shown in Fig. 2(right).

While our method considers both unscattered and scattered components to estimate the parameter set Θ , we focus on the unscattered components in this paper, and the scattered components are described in the supplemental material.

The longitudinal function $M_p(\theta_i, \theta_r)$ is calculated as a normalized Gaussian function $G(\theta_r; -\theta_i + \alpha_p, \beta_{m,p})$ using the following equation:

$$G(\theta_r; -\theta_i + \alpha_p, \beta_{m,p}) = \frac{1}{\sqrt{2\pi}\beta_{m,p}} \exp\left(-\frac{(\theta_r + \theta_i - \alpha_p)^2}{2\beta_{m,p}^2}\right), \quad (1)$$

where α_p and $\beta_{m,p}$ are the slope in the cuticle layer at path p and the parameter representing the variance, respectively.

The azimuthal function $N_p(h, \phi)$ is calculated by the product of the attenuation term A_p and the distribution term D_p .

$$N_p(h, \phi) = A_p(h) D_p(h, \phi), \quad (2)$$

The attenuation term $A_p(h)$ represents the amount of attenuation of light energy along the path p of light incident on h . A_p is calculated from the Fresnel reflectance and transmittance at the boundary of each layer and the attenuation of light in the cortices layer. The distribution term D_p represents how much of the incident light is distributed on the rough surface of the cuticle layer and is expressed by the following normalized Gaussian function

$$D_p(h, \phi) = G(\Phi_p(h) - \phi; 0, \beta_{n,p}), \quad (3)$$

where $\Phi_p(h)$ is the exiting azimuthal angle of light along path p , and $\beta_{n,p}$ is the accumulated azimuthal roughness.

2 PROPOSED METHOD

2.1 Differentiable Rendering of Hair and Fur

The intensity I_q of pixel q is estimated by the following one-sample estimator $\hat{I}$:

$$\hat{I}(\Theta) = L_e(x_0) \prod_{j=1}^{k-1} \frac{S(x_j) \cos^2 \theta_i}{p(x_j)}, \quad (4)$$

where x_j is the j -th vertex of sampled path $\mathbf{x} = x_0 \cdots x_k$ (x_0 is on a light source and x_k is on the camera), L_e is the emittance of the light source, $S(x_j)$ is the BCSDf at j -th vertex on a fiber, p is the probability density function (PDF), and Θ is the set of

optical parameters. The derivative of I_q with respect to the optical parameters Θ is estimated by deriving $\hat{I}_q$ with Θ as:

$$\frac{\partial \hat{I}_q(\Theta)}{\partial \Theta} = L_e(x_0) \sum_{l=1}^{k-1} \frac{\partial S(x_l)}{\partial \Theta} \prod_{j,j \neq l}^{k-1} \frac{S(x_j) \cos^2 \theta_i}{p(x_j)}. \quad (5)$$

We define the loss function $\mathcal{L}$ as:

$$\mathcal{L} = \sum_{q \in \Omega} (I_q - \hat{I}_q(\Theta))^2, \quad (6)$$

where I_q is the pixel intensity of the target image, Ω is the set of pixels covering fibers. The parameter set Θ is obtained by using stochastic gradient descent (SGD) that requires the gradient of $\mathcal{L}$ as:

$$\nabla \mathcal{L} = -2 \sum_{q \in \Omega} (I_q - \hat{I}_q(\Theta)) \frac{\partial \hat{I}_q(\Theta)}{\partial \Theta}. \quad (7)$$

Our method estimates $\hat{I}_q$ and $\frac{\partial \hat{I}_q}{\partial \Theta}$ using path tracing. Since the computational cost of path tracing is costly, our method randomly selects a set of pixels from Ω as a mini-batch of SGD instead of using all the pixels in Ω .

Based on our experiments, we found that the joint estimation of nine parameters in Θ can get stuck in a local minimum. Therefore, we first estimate two optical parameters $\sigma_{c,a}$ and η that affect the entire appearance of hair and fur ($\sigma_{c,a}$ controls the fiber color and η affects the length of light traveled within the fiber, leading to affect the fiber color). Then all the parameters are jointly estimated.

2.2 Derivatives of Unscattered Lobes

2.2.1 Longitudinal function M_p . Since the longitudinal unscattered lobe M_p depends only on the scale tilt for cuticle α and the roughness of cuticle β_m , derivatives of M_p with respect to other parameters are zero.

$$\frac{\partial M_p}{\partial \alpha} = \frac{\partial \alpha_p}{\partial \alpha} \frac{(\theta_r - (-\theta_i + \alpha_p))}{\beta_{m,p}^2} M_p \quad (8)$$

$$\frac{\partial M_p}{\partial \beta_m} = \frac{\partial \beta_{m,p}}{\partial \beta_m} \frac{(\theta_r - (-\theta_i + \alpha_p))^2 - \beta_{m,p}^2}{\beta_{m,p}^3} M_p, \quad (9)$$

where α_p and $\beta_{m,p}$ depend on the path p and the derivatives of α_p and $\beta_{m,p}$ are shown in the supplemental material.

2.2.2 Attenuation term A_p . The attenuation term A_p for unscattered lobes $p \in \mathcal{P}^u$ is represented by:

$$A_R(h) = F, \\ A_p(h) = (1 - F)^2 F^{p-1} T_c T_m,$$

where $p = 1$ and $p = 2$ are used for TT and TRT , respectively. T_c and T_m are the attenuation of light within the cortex and the medulla, respectively.

$$T_c = \exp\left(-\frac{2ps_c\sigma_{c,a}}{\cos \theta_i}\right), \\ T_m = \exp\left(-\frac{2ps_m(\sigma_{m,a} + \sigma_{m,s})}{\cos \theta_i}\right),$$

s_c and s_m represent the (half) distance of light travelled within the cortex and the medulla, respectively as shown in Figure 2.

$$s_c = \sqrt{1 - \sin^2 \gamma_t} - s_m, \\ s_m = \sqrt{\kappa^2 - \sin^2 \gamma_t},$$

where γ_t is the refracted angle in the azimuthal cross section as shown in Fig. 2(right).

The attenuation term A_p depends on five parameters η , l , $\sigma_{c,a}$, $\sigma_{m,s}$, and $\sigma_{m,a}$, and A_p can be derived analytically with respect to these five parameters. To simplify the notation, the derivatives of F , T_c , and T_m with respect to η are represented with $F'(\eta)$, $T'_c(\eta)$, and $T'_m(\eta)$.

$$\frac{\partial A_R}{\partial \eta} = F'(\eta), \quad (10)$$

$$\frac{\partial A_p}{\partial \eta} = (1 - F)F^{p-2}(F'(\eta)T_c T_m((p-1) - (p+1)F) + (1 - F)F(T'_c(\eta)T_m + T_c T'_m(\eta))). \quad (11)$$

Since the derivative of the Fresnel term $F'(\eta)$ is complicated, it is listed in the supplemental material. The derivatives $T'_c(\eta)$ and $T'_m(\eta)$ are calculated by the following equations.

$$T'_c(\eta) = \frac{\partial T_c}{\partial \eta} = -\frac{2p\sigma_{c,a}}{\cos \theta_i} T_c \frac{\partial s_c}{\partial \eta}, \quad (12)$$

$$T'_m(\eta) = \frac{\partial T_m}{\partial \eta} = -\frac{2p(\sigma_{m,a} + \sigma_{m,s})}{\cos \theta_i} T_m \frac{\partial s_m}{\partial \eta}. \quad (13)$$

The derivatives $\frac{\partial s_c}{\partial \eta}$ and $\frac{\partial s_m}{\partial \eta}$ are calculated by:

$$\frac{\partial s_c}{\partial \eta} = \frac{h^2}{\bar{\eta}^3} \left(\frac{1}{\sqrt{1 - \sin^2 \gamma_t}} - \frac{1}{\sqrt{\kappa^2 - \sin^2 \gamma_t}} \right) \frac{\partial \bar{\eta}}{\partial \eta}, \quad (14)$$

$$\frac{\partial s_m}{\partial \eta} = \frac{h^2}{\bar{\eta}^3 \sqrt{\kappa^2 - \sin^2 \gamma_t}} \frac{\partial \bar{\eta}}{\partial \eta}, \quad (15)$$

where the effective refractive index $\bar{\eta}$ and its derivative $\frac{\partial \bar{\eta}}{\partial \eta}$ are calculated as:

$$\bar{\eta} = \frac{\sqrt{\eta^2 - \sin^2 \theta_i}}{\cos \theta_i}, \quad (16)$$

$$\frac{\partial \bar{\eta}}{\partial \eta} = \frac{\eta}{\cos \theta_i \sqrt{\eta^2 - \sin^2 \theta_i}}. \quad (17)$$

The derivatives of the attenuation term A_p with respect to the layers of cuticle l are calculated by the following equations:

$$\frac{\partial A_R}{\partial l} = F'(l), \quad (18)$$

$$\frac{\partial A_p}{\partial l} = F'(l)(1 - F)F^{p-2}((p-1) - (p+1)F)T_c T_m, \quad (19)$$

where the derivative $F'(l) = \frac{\partial F}{\partial l}$ is shown in the supplemental material.

Next, we describe the derivatives of the attenuation term A_p with respect to $\sigma_{c,a}$, $\sigma_{m,s}$, and $\sigma_{m,a}$. Since the attenuation term A_R for lobe $p = R$ does not depend on these parameters, the derivatives of A_R for these parameters are zero. The derivatives of the

Table 2: Estimated values of the optical parameter set Θ using our method. The values in the parentheses indicate the errors between the estimated values and the true values used to render the target images.

	α	β_m	β_n	η	l	$\sigma_{c,a}$	$\sigma_{m,s}$	$\sigma_{m,a}$	g
Fig. 1(top)	1.19	2.70	4.94	1.22	1.54	1.48, 1.76, 2.16	2.60	0.28	0.29
	(+0.70)	(+0.65)	(+1.19)	(+0.02)	(-0.25)	(-0.02, -0.04, -0.09)	(-0.89)	(+0.28)	(+0.01)
Fig. 1(bottom)	1.67	2.95	5.07	1.24	1.43	0.25	2.67	0.10	0.24
	(-0.14)	(-4.49)	(-1.81)	(+0.05)	(-0.57)	(0.00)	(+0.37)	(-0.04)	(+0.16)

attenuation term A_p for other lobes $p = \{TT, TRT\}$ are calculated by the following equations.

$$\frac{\partial A_p}{\partial \sigma_{c,a}} = -\frac{2ps_c}{\cos \theta_i} A_p, \quad (20)$$

$$\frac{\partial A_p}{\partial \sigma_{m,s}} = -\frac{2ps_m}{\cos \theta_i} A_p, \quad (21)$$

$$\frac{\partial A_p}{\partial \sigma_{m,a}} = -\frac{2ps_m}{\cos \theta_i} A_p. \quad (22)$$

2.2.3 Derivatives of distribution term D_p . The distribution term D_p for unscattered lobes $\mathcal{P}^u$ depends only on β_n and η . The derivative $\frac{\partial D_p}{\partial \beta_n}$ is calculated by:

$$\frac{\partial D_p}{\partial \beta_n} = \frac{d\beta_{n,p}}{d\beta_n} \frac{((\Phi_p(h) - \phi)^2 - \beta_{n,p}^2)}{\beta_{n,p}^3} D_p, \quad (23)$$

where $\frac{d\beta_{n,p}}{d\beta_n}$ takes different values for each lobe p as shown in the supplemental material.

The derivative of D_p with respect to the refractive index η is calculated by the following equation.

$$\frac{\partial D_p}{\partial \eta} = -\frac{\partial(\Phi_p(h) - \phi)}{\partial \eta} \frac{\Phi_p(h) - \phi}{\beta_{n,p}^2} D_p, \quad (24)$$

$$\frac{\partial \Phi_p(h)}{\partial \eta} = 2p \frac{\partial \gamma_t}{\partial \eta}, \quad (25)$$

$$\frac{\partial \gamma_t}{\partial \eta} = -\frac{h}{\eta^2 \sqrt{1 - \sin^2 \gamma_t}} \frac{\partial \eta}{\partial \eta}. \quad (26)$$

For lobe $p = R$, the derivative $\frac{\partial D_p}{\partial \eta}$ is zero since $\frac{\partial \Phi_R(h)}{\partial \eta} = 0$.

3 RESULTS

Fig. 1 shows the rendering results of our parameter estimation using differentiable rendering. Fig. 1 (top) shows the rendering results of human hair where the initial parameter values are set to reproduce blonde hair, while the target images are brown hair. Fig. 1 (bottom) shows the rendering results of fur (the parameter values of the target image are for raccoon [Yan et al. 2017]). As shown in the rightmost error images, our method can estimate optical parameters that provide rendering results whose appearance matches the target images.

Table 2 shows the estimated parameter values, and the errors between the estimated values and the ground truth used to render the target images. As shown in Table 2, the estimated values are close to the ground truth, especially for η and $\sigma_{c,a}$.

Our method uses 2048 spp (samples per pixel) to estimate the pixel intensity $\hat{I}_q$ and 128 spp to estimate the gradient (Eq. (5)).

The mini-batch size (number of pixels used in path tracing at each iteration) is set to 16, and our method employs up to $k = 5$ bounce paths. The computational times to estimate Θ in Fig. 1 are 65 min (human hair) and 35 min (fur) measured on a standard PC with CPU Intel Core-i9 10850k 3.60GHz.

4 CONCLUSIONS AND FUTURE WORK

We have proposed an optical parameter estimation method of BCSDf using differentiable rendering. We derive the gradients of the longitudinal function M_p , the attenuation term A_p , and the distribution term D_p explicitly for nine optical parameters. Our differentiable path tracing renderer calculates the gradients of the pixel values using the derived gradients. The experiments show that rendering results with our estimated values are similar to the target synthesized images.

Currently, our method imposes on premises that the fiber geometries are given. For future work, to enhance the applicability of our method, we would like to estimate optical parameters of hair and fur from a single photograph where the fiber geometries are not known prior.

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